

Question	Answer	Marks
4(a)	longitudinal waves have oscillations <u>parallel</u> to the (direction of) transfer of energy	B1
	transverse waves have oscillations <u>perpendicular</u> to the (direction of) transfer of energy	B1
4(b)(i)	A marked at the open end of the pipe	B1
4(b)(ii)	$f = v / \lambda$	C1
	$\lambda = 4 \times 4.5 \times 10^{-2}$	C1
	$f = 340 / (4 \times 4.5 \times 10^{-2})$	A1
	= 1900 Hz	
4(b)(iii)	the node–antinode distance is longer or the wavelength (of the wave) is longer	M1
	(the speed of sound is constant so) the frequency (of the wave) is lower	A1

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